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Waveguide angle sensing

Abstract

A head-mounted device may have a projector that emits light. The light may include visible images produced using image data and optionally infrared light. A waveguide may propagate the light and may have a first output coupler that couples the visible images out of the waveguide and towards an eye box. One or more optical sensors may generate optical sensor data based on the light emitted by the projector. Control circuitry may detect an angle between the waveguide and the projector based on the optical sensor data. The control circuitry may adjust the image data to compensate for changes in the angle over time. The optical sensors may include an optical sensor mounted to the projector and facing the waveguide and/or one or more optical sensors mounted to an optical bridge sensor.

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Background/Summary

(1) This application claims the benefit of U.S. Provisional Patent Application No. 63/354,201, filed Jun. 21, 2022, which is hereby incorporated by reference herein in its entirety.

FIELD

(1) This relates generally to electronic devices, and, more particularly, to electronic devices such as head-mounted devices.

BACKGROUND

(2) Electronic devices have components such as displays and other optical components. During operation, there is a risk that components may become misaligned with respect to each other due to drop events and other undesired high-stress events. This poses challenges for ensuring satisfactory component performance.

SUMMARY

- (3) A head-mounted device such as a pair of glasses may have a head-mounted housing. The head-mounted device may include displays such as projector displays and may include associated optical components. The housing may have a first portion, a second portion, and a nose bridge that couples the first portion to the second portion. A first display having a first projector and a first waveguide may be mounted in the first portion of the housing. A second display having a second projector and a second waveguide may be mounted in the second portion of the housing.
- (4) One of the projectors may emit light. The light may include visible light images that are produced by the projector based on image data. One of the waveguides may propagate the light via total internal reflection. The waveguide may have an output coupler that couples the visible light images in the light out of the waveguide and towards an eye box. If desired, the light may also include infrared light. One or more optical sensors (e.g., image sensors) may generate optical sensor data (e.g., image sensor data) based on the light emitted by the projector. Control circuitry may detect an angle between the waveguide and the projector based on the optical sensor data. The control circuitry may adjust the image data to compensate for changes in the angle over time.
- (5) The optical sensors may include an optical sensor mounted to the projector and facing the waveguide. The optical sensor may generate the optical sensor data based on a portion of the light that has reflected towards the optical sensor from the waveguide. The optical sensors may additionally or alternatively include an optical sensor in an optical bridge sensor. In these examples, the waveguide may include an additional output coupler. The additional output coupler may have a reflective diffractive mode that couples some of the light out of the waveguide and towards the optical sensor in the optical bridge sensor. The additional output coupler may also have a transmissive diffractive mode that couples some of the light out of the waveguide and towards a mirror or an additional optical sensor opposite the optical sensor. The mirror may reflect the light towards the optical sensor for generating the optical sensor data or the additional sensor may generate a portion of the optical sensor data based on the light received from the additional output coupler.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram of an illustrative system in accordance with some embodiments.
- (2) FIG. 2 is a top view of an illustrative head-mounted device in accordance with some embodiments.
- (3) FIG. 3 is a top view of an illustrative display projector and waveguide for providing image light to an eye box in accordance with some embodiments.
- (4) FIG. 4 is a top view of an illustrative display projector having an optical sensor for detecting a waveguide angle in accordance with some embodiments.
- (5) FIG. 5 is a top view of an illustrative display projector having an infrared emitter and an infrared optical sensor for detecting a waveguide angle in accordance with some embodiments.
- (6) FIG. 6 is a top view of an illustrative display system having an optical bridge sensor and a mirror for detecting a waveguide angle in accordance with some embodiments.
- (7) FIG. 7 is a top view of an illustrative display system having optical sensors at opposing sides of a waveguide for detecting a waveguide angle in accordance with some embodiments.
- (8) FIG. 8 is a flow chart of illustrative operations involved in using a system of the type shown in FIGS. 1-8 to detect and mitigate changes in waveguide angle in accordance with some embodiments.

DETAILED DESCRIPTION

- (9) A system may include one or more electronic devices. Each device may contain optical components and other components. FIG. 1 is a schematic diagram of an illustrative system of the

type that may include one or more electronic devices with optical components. As shown in FIG. 1, system **8** may include electronic devices **10**. Devices **10** may include head-mounted devices (e.g., goggles, glasses, helmets, and/or other head-mounted devices), cellular telephones, tablet computers, peripheral devices such as headphones, game controllers, and/or other input devices. Devices **10** may, if desired, include laptop computers, computer monitors containing embedded computers, desktop computers, media players, or other handheld or portable electronic devices, smaller devices such as wristwatch devices, pendant devices, ear buds, or other wearable or miniature devices, televisions, computer displays that do not contain embedded computers, gaming devices, remote controls, embedded systems such as systems in which equipment is mounted in a kiosk, in an automobile, airplane, or other vehicle, removable external cases for electronic equipment, straps, wrist bands or head bands, removable covers for electronic devices, cases or bags that receive and carry electronic equipment and other items, necklaces or arm bands, wallets, sleeves, pockets, or other structures into which electronic equipment or other items may be inserted, part of an item of clothing or other wearable item (e.g., a hat, belt, wrist band, headband, sock, glove, shirt, pants, etc.), or equipment that implements the functionality of two or more of these devices.

(10) With one illustrative configuration, which may sometimes be described herein as an example, system **8** includes a head-mounted device such as a pair of glasses (sometimes referred to as augmented reality glasses). System **8** may also include peripherals such as headphones, game controllers, and/or other input-output devices (as examples). In some scenarios, system **8** may include one or more stand-alone devices **10**. In other scenarios, multiple devices **10** in system **8** exchange information using wired and/or wireless links, which allows these devices **10** to be used together. For example, a first of devices **10** may gather user input or other input that is used to control a second of devices **10** (e.g., the first device may be a controller for the second device). As another example, a first of devices **10** may gather input that is used in controlling a second device **10** that, in turn, displays content on a third device **10**.

(11) Devices **10** may include components **12**. Components **12** may include control circuitry. The control circuitry may include storage and processing circuitry for supporting the operation of system **8**. The storage and processing circuitry may include storage such as nonvolatile memory (e.g., flash memory or other electrically-programmable-read-only memory configured to form a solid state drive), volatile memory (e.g., static or dynamic random-access-memory), etc. Processing circuitry in the control circuitry may be used to gather input from sensors and other input devices and may be used to control output devices. The processing circuitry may be based on one or more microprocessors, microcontrollers, digital signal processors, baseband processors and other wireless communications circuits, power management units, audio chips, application specific integrated circuits, etc.

(12) To support communications between devices **10** and/or to support communications between equipment in system **8** and external electronic equipment, devices **10** may include wired and/or wireless communications circuitry. The communications circuitry of devices **10**, which may sometimes be referred to as control circuitry and/or control and communications circuitry, may include antennas, radio-frequency transceiver circuitry, and other wireless communications circuitry and/or wired communications circuitry. The communications circuitry of devices **10** may, for example, support bidirectional wireless communications between devices **10** over wireless links such as wireless link **14** (e.g., a wireless local area network link, a near-field communications link, or other suitable wired or wireless communications link (e.g., a Bluetooth® link, a WiFi® link, a 60 GHz link or other millimeter wave link, etc.)). Components **12** may also include power circuits for transmitting and/or receiving wired and/or wireless power and may include batteries.

(13) Components **12** may include input-output devices. The input-output devices may be used in gathering user input, in gathering information on the environment surrounding the user, and/or in providing a user with output. The input-output devices may include sensors such as force sensors

(e.g., strain gauges, capacitive force sensors, resistive force sensors, etc.), audio sensors such as microphones, touch and/or proximity sensors such as capacitive sensors, optical sensors such as optical sensors that emit and detect light, ultrasonic sensors, and/or other touch sensors and/or proximity sensors, monochromatic and color ambient light sensors, optical sensor (e.g., image sensors), sensors for detecting position, orientation, and/or motion (e.g., accelerometers, magnetic sensors such as compass sensors, gyroscopes, and/or inertial measurement units that contain some or all of these sensors), radio-frequency sensors, depth sensors (e.g., structured light sensors and/or depth sensors based on stereo imaging devices), optical sensors such as self-mixing sensors and light detection and ranging (lidar) sensors that gather time-of-flight measurements, humidity sensors, moisture sensors, and/or other sensors. In some arrangements, devices **10** may use sensors and/or other input-output devices to gather user input (e.g., buttons may be used to gather button press input, touch sensors overlapping displays can be used for gathering user touch screen input, touch pads may be used in gathering touch input, microphones may be used for gathering audio input, accelerometers may be used in monitoring when a finger contacts an input surface and may therefore be used to gather finger press input, etc.).

(14) Components **12** may include haptic output devices. The haptic output devices can produce motion that is sensed by the user (e.g., through the user's head, hands, or other body parts). Haptic output devices may include actuators such as electromagnetic actuators, motors, piezoelectric actuators, electroactive polymer actuators, vibrators, linear actuators, rotational actuators, actuators that bend bendable members, etc.

(15) If desired, input-output devices in components **12** may include other devices such as displays (e.g., to display images for a user), status indicator lights (e.g., a light-emitting diode that serves as a power indicator, and other light-based output devices), speakers and other audio output devices, electromagnets, permanent magnets, structures formed from magnetic material (e.g., iron bars or other ferromagnetic members that are attracted to magnets such as electromagnets and/or permanent magnets), etc.

(16) As shown in FIG. **1**, sensors such as position sensors **16** may be mounted to one or more of components **12**. Position sensors **16** may include optical sensors (e.g., visible or infrared optical sensors), accelerometers, magnetic sensors such as compass sensors, gyroscopes, and/or inertial measurement units (IMUs) that contain some or all of these sensors. Position sensors **16** may be used to measure location (e.g., location along X, Y, and Z axes), orientation (e.g., angular orientation around the X, Y, and Z axes), and/or motion (changes in location and/or orientation as a function of time). Sensors such as position sensors **16** that can measure location, orientation, and/or motion may sometimes be referred to herein as position sensors, motion sensors, and/or orientation sensors.

(17) Devices **10** may use position sensors **16** to monitor the position (e.g., location, orientation, motion, etc.) of devices **10** in real time. This information may be used in controlling one or more devices **10** in system **8**. As an example, a user may use a first of devices **10** as a controller. By changing the position of the first device, the user may control a second of devices **10** (or a third of devices **10** that operates in conjunction with a second of devices **10**). As an example, a first device may be used as a game controller that supplies user commands to a second device that is displaying an interactive game.

(18) Devices **10** may also use position sensors **16** to detect any changes in position of components **12** with respect to the housings and other structures of devices **10** and/or with respect to each other. For example, a given one of devices **10** may use a first position sensor **16** to measure the position of a first of components **12**, may use a second position sensor **16** to measure the position of a second of components **12**, and may use a third position sensor **16** to measure the position of a third of components **12**. By comparing the measured positions of the first, second, and third components (and/or by using additional sensor data), device **10** can determine whether calibration operations should be performed, how calibration operations should be performed, and/or when/how other

operations in device **10** should be performed.

(19) In an illustrative configuration, devices **10** include a head-mounted device such as a pair of glasses (sometimes referred to as augmented reality glasses). A top view of device **10** in an illustrative configuration in which device **10** is a pair of glasses is shown in FIG. **2**. A shown in FIG. **2**, device **10** may include housing **18**. Housing **18** may include a main portion (sometimes referred to as a glasses frame) such as main portion **18M** and temples **18T** that are coupled to main portion **18M** by hinges **18H**. Nose bridge portion NB may have a recess that allows housing **18** to rest on a nose of a user while temples **18T** rest on the user's ears.

(20) Images may be displayed in eye boxes **20** using displays **22** and waveguides **24**. Displays **22** may sometimes be referred to herein as projectors **22**, projector displays **22**, display projectors **22**, light projectors **22**, image projectors **22**, light engines **22**, or display modules **22**. Projector **22** may include a first projector **22B** (sometimes referred to herein as left projector **22B**) and a second projector **22A** (sometimes referred to herein as right projector **22A**). Projectors **22A** and **22B** may be mounted at opposing right and left edges of main portion **18M** of housing **18**, for example. Eye boxes **20** may include a first eye box **20B** (sometimes referred to herein as left eye box **20B**) and may include a second eye box **20A** (sometimes referred to herein as right eye box **20A**).

Waveguides **24** may include a first waveguide **24B** (sometimes referred to herein as left waveguide **24B**) and a second waveguide **24A** (sometimes referred to herein as right waveguide **24A**). Main portion **18M** of housing **18** may, for example, have a first portion that includes projector **22B** and waveguide **24B** and a second portion that includes projector **22A** and waveguide **24A** (e.g., where nose bridge NB separates the first and second portions such that the first portion is at a first side of the nose bridge and the second portion is at a second side of the nose bridge).

(21) Waveguides **24** may each include one or more stacked substrates (e.g., stacked planar and/or curved layers sometimes referred to herein as waveguide substrates) of optically transparent material such as plastic, polymer, glass, etc. If desired, waveguides **24** may also include one or more layers of holographic recording media (sometimes referred to herein as holographic media, grating media, or diffraction grating media) on which one or more diffractive gratings are recorded (e.g., holographic phase gratings, sometimes referred to herein as holograms). A holographic recording may be stored as an optical interference pattern (e.g., alternating regions of different indices of refraction) within a photosensitive optical material such as the holographic media. The optical interference pattern may create a holographic phase grating that, when illuminated with a given light source, diffracts light to create a three-dimensional reconstruction of the holographic recording. The holographic phase grating may be a non-switchable diffractive grating that is encoded with a permanent interference pattern or may be a switchable diffractive grating in which the diffracted light can be modulated by controlling an electric field applied to the holographic recording medium. Multiple holographic phase gratings (holograms) may be recorded within (e.g., superimposed within) the same volume of holographic medium if desired. The holographic phase gratings may be, for example, volume holograms or thin-film holograms in the grating medium. The grating media may include photopolymers, gelatin such as dichromated gelatin, silver halides, holographic polymer dispersed liquid crystal, or other suitable holographic media.

(22) Diffractive gratings on waveguides **24** may include holographic phase gratings such as volume holograms or thin-film holograms, meta-gratings, or any other desired diffractive grating structures. The diffractive gratings on waveguides **24** may also include surface relief gratings (SRGs) formed on one or more surfaces of the substrates in waveguides **24**, gratings formed from patterns of metal structures, etc. The diffractive gratings may, for example, include multiple multiplexed gratings (e.g., holograms) that at least partially overlap within the same volume of grating medium (e.g., for diffracting different colors of light and/or light from a range of different input angles at one or more corresponding output angles).

(23) Waveguides **24** may have input couplers that receive light from projectors **22**. This image light is then guided laterally (along the X axis) within waveguides **24** in accordance with the principal of

total internal reflection. Each waveguide **24** may have an output coupler in front of a respective eye box **20**. The output coupler couples the image light out of the waveguide **24** and directs an image towards the associated eye box **20** for viewing by a user (e.g., a user whose eyes are located in eye boxes **20**), as shown by arrows **26**. Input and output couplers for device **10** may be formed from diffractive gratings (e.g., surface relief gratings, volume holograms, etc.) and/or other optical structures.

(24) For example, as shown in FIG. 2, projector **22B** may emit (e.g., produce, generate, project, or display) image light that is coupled into waveguide **24B** (e.g., by a first input coupler on waveguide **24B**). The image light may propagate in the +X direction along waveguide **24B** via total internal reflection. The output coupler on waveguide **24B** may couple the image light out of waveguide **24B** and towards eye box **20B** (e.g., for view by the user's left eye at eye box **20B**). Similarly, projector **22A** may emit (e.g., produce, generate, project, or display) image light that is coupled into waveguide **24A** (e.g., by a second input coupler on waveguide **24A**). The image light may propagate in the -X direction along waveguide **24A** via total internal reflection. The output coupler on waveguide **24A** may couple the image light out of waveguide **24A** and towards eye box **20A** (e.g., for view by the viewer's right eye at eye box **20A**).

(25) FIG. 3 is a top view showing how waveguide **24B** may provide light to eye box **20B**. As shown in FIG. 3, projector **22B** may emit image light **38B** that is provided to waveguide **24B**. Projector **22B** may include collimating optics (sometimes referred to as an eyepiece, eyepiece lens, or collimating lens) that help direct image light **38B** towards waveguide **24B**. Projector **22B** may generate image light **38B** associated with image content to be displayed to (at) eye box **20B**. Projector **22B** may include light sources that produce image light **38B** (e.g., in scenarios where projector **22B** is an emissive display module, the light sources may include arrays of light emitters such as LEDs) or may include light sources that produce illumination light that is provided to a spatial light modulator in projector **22B**. The spatial light modulator may modulate the illumination light with (using) image data (e.g., a series of image frames) to produce image light **38B** (e.g., image light that includes images as identified by the image data). The spatial light modulator may be a transmissive spatial light modulator (e.g., may include a transmissive display panel such as a transmissive LCD panel) or a reflective spatial light modulator (e.g., may include a reflective display panel such as a DMD display panel, an LCOS display panel, an fLCOS display panel, etc.).

(26) Waveguide **24B** may be used to present image light **38B** output from projector **22B** to eye box **20B**. Waveguide **24B** may include one or more optical couplers such as input coupler **28B**, cross-coupler **32B**, and output coupler **30B**. In the example of FIG. 3, input coupler **28B**, cross-coupler **32B**, and output coupler **30B** are formed at or on waveguide **24B**. Input coupler **28B**, cross-coupler **32B**, and/or output coupler **30B** may be completely embedded within the substrate layers of waveguide **24B**, may be partially embedded within the substrate layers of waveguide **24B**, may be mounted to waveguide **24B** (e.g., mounted to an exterior surface of waveguide **24B**), etc.

(27) The example of FIG. 3 is merely illustrative. One or more of these couplers (e.g., cross-coupler **32B**) may be omitted. Waveguide **24B** may be replaced with multiple waveguides that are laterally and/or vertically stacked with respect to each other. Each of these waveguides may include one, two, all, or none of couplers **28B**, **32B**, and **30B**. Waveguide **24B** may be at least partially curved or bent if desired.

(28) Waveguide **24B** may guide image light **38B** down its length via total internal reflection. Input coupler **28B** may be configured to couple image light **38B** into waveguide **24B**, whereas output coupler **30B** may be configured to couple image light **38B** from within waveguide **24B** to the exterior of waveguide **24B** and towards eye box **20B**. Input coupler **28B** may include an input coupling prism, one or more mirrors (e.g., louvered partially reflective mirrors), or diffractive gratings such as an SRG or a set of volume holograms, as examples.

(29) As shown in FIG. 3, projector **22B** may emit image light **38B** in the +Y direction towards waveguide **24B**. When image light **38B** strikes input coupler **28B**, input coupler **28B** may redirect

image light **38B** so that the light propagates within waveguide **24B** via total internal reflection towards output coupler **30B** (e.g., in the +X direction). When image light **38B** strikes output coupler **30B**, output coupler **30B** may redirect image light **38B** out of waveguide **24B** towards eye box **20B** (e.g., back in the -Y direction). In scenarios where cross-coupler **32B** is formed at waveguide **24B**, cross-coupler **32B** may redirect image light **38B** in one or more directions as it propagates down the length of waveguide **24B**, for example. Cross-coupler **32B** may expand a pupil of image light **38B** if desired.

(30) Input coupler **28B**, cross-coupler **32B**, and/or output coupler **30B** may be based on reflective and refractive optics or may be based on holographic (e.g., diffractive) optics. In arrangements where couplers **28B**, **30B**, and **32B** are formed from reflective and refractive optics, couplers **28B**, **30B**, and **32B** may include one or more reflectors (e.g., an array of micromirrors, partial mirrors, louvered mirrors, or other reflectors). In arrangements where couplers **28B**, **30B**, and **32B** are based on holographic optics, couplers **28B**, **30B**, and **32B** may include diffractive gratings (e.g., volume holograms, surface relief gratings, etc.). Any desired combination of holographic and reflective optics may be used to form couplers **28B**, **30B**, and **32B**. In one suitable arrangement that is sometimes described herein as an example, input coupler **28B**, cross-coupler **32B**, and output coupler **30B** each include surface relief gratings (e.g., surface relief gratings formed by modulating the thickness of one or more layers of surface relief grating substrate in waveguide **24B**).

(31) In an augmented reality configuration, waveguide **24B** may also transmit (pass) real-world light from the scene/environment in front of (facing) device **10**. The real-world light (sometimes referred to herein as world light or environmental light) may include light emitted and/or reflected by objects in the scene/environment in front of device **10**. For example, output coupler **30B** may transmit world light **36** from real-world objects **34** in the scene/environment in front of device **10**. Output coupler **30B** may, for example, diffract image light **38B** to couple image light **38B** out of waveguide **24B** and towards eye box **20B** while transmitting world light **36** (e.g., without diffracting world light **36**) to eye box **20B**. This may allow images in image light **38B** to be overlaid with world light **36** of real-world objects **34** (e.g., to overlay virtual objects from image data in image light **38B** as displayed by projector **22B** with real-world objects **34** in front of the user when viewed at eye box **20A**).

(32) In the example of FIG. 3, only the waveguide and projector for providing image light to eye box **20B** is shown for the sake of clarity. Waveguide **24A** (FIG. 2) may include similar structures for providing light to eye box **20A**. During operation of device **10** (e.g., by an end user), mechanical stresses, thermal effects, and other stressors may alter the alignment between two or more components of device **10**.

(33) For example, the optical alignment between the components of device **10** may change when the user places device **10** on their head, removes device **10** from their head, places device **10** on a surface or within a case, or drops device **10** on the ground, when a mechanical impact event occurs at device **10**, when device **10** enters different environments at different temperatures or humidities, when a user bends, stresses, or shakes one or more components in device **10**, etc. If care is not taken, these changes in optical alignment can undesirably affect the images provided to eye boxes **20A** and/or **20B** (e.g., can produce visible misalignment or distortion at one or both eye boxes **20A** and **20B**). As these changes in optical alignment will vary by user and from system-to-system, it may be desirable to actively identify such changes in the field (e.g., during operation of device **10** by an end user rather than in-factory during the manufacture of device **10**) so that suitable action can be taken to mitigate the identified changes to provide an optimal display experience for the user over time.

(34) Device **10** may perform in-field calibration operations to detect and optionally mitigate optical misalignment using a set of sensors. The set of one or more sensors may measure a waveguide angle between projector **22B** and waveguide **24B**. Changes in the waveguide angle over time may be associated with optical misalignment between the images displayed by the projector and the

images as viewed at the eye box and/or may be associated with optical misalignment between the left and right eye boxes. By measuring (detecting) the waveguide angle, the set of sensors may mitigate such optical misalignment. If desired, the set of sensors that measure waveguide angle may include an optical sensor mounted to projector **22B**.

(35) FIG. **4** is a top view showing how device **10** may include an optical sensor mounted to projector **22B** for measuring waveguide angle. As shown in FIG. **4**, projector **22B** may be optically coupled to a first (left) edge of waveguide **24B** (e.g., a temple side/edge of the first waveguide). Projector **22B** may emit image light **38B**. Input coupler **28B** may couple image light **38B** into waveguide **24B** (e.g., within the total internal reflection (TIR) range of the waveguide). Waveguide **24B** may propagate image light **38B** along its length via total internal reflection. Cross-coupler **32B** (FIG. **3**) and output coupler **30B** have been omitted from FIG. **4** for the sake of clarity.

(36) Waveguide **24B** may have a lateral surface **48** (sometimes referred to herein as a lateral waveguide surface of a simply as a waveguide surface). Lateral surface **48** may face the user or, if desired, may face world-side. Projector **22B** may also have a lateral (output) surface **50** facing waveguide **24B**. Lateral surface **48** may lie within plane **42**. Lateral surface **50** may lie within plane **44**. Plane **42** (lateral surface **48** of waveguide **24B**) may be oriented at a waveguide angle θ with respect to plane **44** (lateral surface **50** of waveguide **24B**). In other words, waveguide **24B** may be tilted or oriented at waveguide angle θ with respect to projector **22B**. Over time, waveguide angle θ may change from a nominal, ideal, or expected waveguide angle (e.g., zero degrees) due to mechanical stresses on device **10**. This may cause optical misalignment in device **10**.

(37) An optical sensor such as optical sensor **40** may be mounted on or in projector **22B** for measuring waveguide angle θ . Optical sensor **40** may be mounted to lateral surface **50** of projector **22B**, as one example. Optical sensor **40** of FIG. **4** may be used to detect (measure) waveguide angle θ based on the image light **38B** emitted by projector **22B**. Optical sensor **40** may include one or more optical sensors (e.g., one or more visible light optical sensors or cameras having one or more optical sensor pixels arranged in corresponding array(s) of optical sensor pixels). One or more lenses or microlenses may be disposed overlapping the optical sensor(s). Optical sensor **40** may include a quad cell sensor as one example.

(38) Optical sensor **40** may receive a portion **38B'** of the image light **38B** emitted by projector **22B**. For example, some of image light **38B** (i.e., portion **38B'**) may reflect off lateral surface **48** and/or the opposing lateral surface of waveguide **24B** and towards optical sensor **40** (e.g., outside of the TIR range of the waveguide). Some of image light **38B** (i.e., portion **38B'**) may additionally or alternatively be diffracted towards optical sensor **40** by input coupler **28B** (e.g., at angles outside the TIR range of the waveguide). For example, in implementations where input coupler **28B** includes a diffractive grating, input coupler **28B** may reflect some portion **38B'** of image light **38B** towards optical sensor **40** via zero-order reflection of portion **38B'** by the diffractive grating.

(39) The portion **38B'** of image light **38B** received at optical sensor **40** may be used to detect waveguide angle θ . For example, optical sensor **40** may generate optical sensor data (e.g., image sensor data) based on portion **38B'** of the image light **38B**. Different regions of optical sensor **40** (e.g., different sensor pixels or cells) will be illuminated by portion **38B'** when different waveguide angles θ are present between waveguide **24B** and projector **22B**. Control circuitry (e.g., one or more processors) may use the optical sensor data to identify (e.g., compute, calculate, generate, measure, detect, etc.) waveguide angle θ by, for example, comparing the regions of optical sensor **40** that are illuminated by portion **38B'** (or the intensity of measured light at different sensor pixel positions) to predetermined (e.g., expected, calibrated, ideal, nominal, etc.) regions of optical sensor **40** (or expected optical sensor data) that would be measured when waveguide angle θ has a nominal value (e.g., zero degrees). Once waveguide angle θ is known, the one or more processors may mechanically adjust the optical alignment of one or more components in device **10** and/or may adjust the image data used to produce image light **38B** (e.g., by pre-distorting and/or color-compensate the image light) in a manner that mitigates any optical misalignment or non-idealities

produced by the detected waveguide angle θ differing from the expected or nominal waveguide angle θ .

(40) Image light **38B** and thus portion **38B'** of image light **38B** include visible light (e.g., for display at the eye box). The example of FIG. **4** in which optical sensor **40** detects waveguide angle θ based on portion **38B'** of image light **38B** is merely illustrative. If desired, optical sensor **40** may detect waveguide angle θ based on infrared light emitted by projector **22B**.

(41) FIG. **5** is a top view showing one example of how projector **22B** may emit infrared light for measurement by optical sensor **40**. As shown in FIG. **5**, projector **22B** may include illumination optics **52**. Illumination optics **52** may include one or more light sources that emit illumination light **66** in one or more visible wavelength ranges (e.g., red, green, and blue illumination light).

(42) Projector **22B** may include a spatial light modulator such as reflective spatial light modulator **54**. Reflective spatial light modulator **54** may include a reflective display panel such as a digital micromirror device (DMD) panel, a liquid crystal on silicon (LCOS) panel, a ferroelectric liquid crystal on silicon (fLCOS) panel, or other spatial light modulators. Optics **56** (e.g., one or more optical wedges or prisms, partial reflectors, polarizers, reflective polarizers, or other structures) may direct illumination light **66** to reflective spatial light modulator **54**.

(43) Reflective spatial light modulator **54** may be controlled using image data to selectively reflect illumination light **66** at different pixel positions (e.g., as determined by the image data) to produce image light **38B**. In other words, reflective spatial light modulator **54** may modulate image data onto illumination light **66** to produce image light **38B** or may modulate illumination light **66** using the image data to produce image light **38B**. Optics **56** may redirect image light **38B** towards input coupler **28B** on waveguide **24B**. Collimating optics **58** (e.g., one or more lens elements sometimes referred to as eyepiece optics) in projector **22B** may help to direct and collimate image light **38B** towards input coupler **28B**.

(44) Projector **22B** may include one or more infrared emitters such as infrared emitter **60**. Infrared emitter **60**, optics **56**, illumination optics **52**, reflective spatial light modulator **54**, and/or collimating optics **58** may be disposed in a housing for projector **22B** if desired. Infrared emitter **60** may emit infrared light **64**, which is directed towards reflective spatial light modulator **54** by lens **62** and optics **56**. The infrared light referred to herein may include infrared (IR) and/or near infrared (NIR) wavelengths. Reflective display panel **54** may reflect infrared light **64**. Optics **56** and collimating optics **58** may direct infrared light **64** towards waveguide **24B**. A lateral surface of waveguide **24B** and/or a zero-order mode of a diffractive grating in input coupler **28B** may reflect a portion **64'** of infrared light **64** towards optical sensor **40**. Optical sensor **40** may generate optical sensor data (e.g., infrared optical sensor data) based on the portion **64'** of infrared light **64**, which is then used to detect waveguide angle θ . If desired, input coupler **28B** may couple an additional portion of infrared light **64** into waveguide **24B** (e.g., within the TIR range of the waveguide). This portion of infrared light **64** may propagate down waveguide **24B** and may be coupled out of waveguide **24B** (e.g., by output coupler **30B** of FIG. **3**) and directed towards the eye box. This portion of infrared light **64** may reflect off the user's eye at the eye box and may be detected by an infrared optical sensor in device **10** to detect and track the direction of the user's gaze over time (e.g., infrared light **64** may also be used to perform gaze tracking).

(45) If desired, an optical bridge sensor for device **10** may be used to detect waveguide angle θ . FIG. **6** is a top view showing one example of how an optical bridge sensor for device **10** may be used to detect waveguide angle θ . As shown in FIG. **6**, projector **22B** may emit image light **38B** that is coupled into waveguide **24B** by input coupler **28B**. Waveguide **24B** may be tilted or oriented at waveguide angle θ with respect to projector **22B**. Waveguide **24B** may propagate image light **38B** along its length. Output coupler **30B** may couple some of image light **38B** out of waveguide **24B** and towards eye box **20B** (in the direction of arrow **70**).

(46) As shown in FIG. **6**, an optical sensor such as optical bridge sensor **74** may be disposed in device **10** at an opposite end of waveguide **24B** from projector **22B**. For example, optical bridge

sensor **74** may be disposed within nose bridge NB of main portion **18M** of the housing (FIG. 2). Optical bridge sensor **74** may be optically coupled to waveguides **24B** and **24A** (FIG. 2) and may, if desired, be mounted to waveguides **24A** and **24B** (e.g., using a mounting bracket or frame). Waveguide **24B** may include an additional output coupler at the end of waveguide **24B** opposite projector **22B** such as output coupler **77B** (sometimes referred to herein as supplemental output coupler **77B**, bridge output coupler **77B**, or bridge sensor output coupler **77B**). Output coupler **77B** may couple some of the image light **38B** propagating through waveguide **24B** (e.g., a portion of image light **38B** not coupled out of the waveguide by output coupler **30B**) out of waveguide **24B** and into optical bridge sensor **74** (in the direction of arrow **72**). Similarly, waveguide **24A** may include an additional output coupler that couples some of the image light propagating through waveguide **24A** out of waveguide **24A** and into optical bridge sensor **74**.

(47) Optical bridge sensor **74** may include one or more optical sensors **78** that gather (e.g., generate, capture, detect, measure, produce, etc.) optical sensor data (sometimes referred to herein as optical bridge sensor data) from the image light coupled out of waveguides **24A** and/or **24B**. Optical sensor(s) **78** may be mounted to a common package or substrate such as substrate **71** (e.g., a rigid or flexible printed circuit board). The optical sensor data gathered by optical bridge sensor **74** may be a real-time representation of the image data that is actually being provided to eye box **20B** after propagating from the projector **22B** and through waveguide **24B**. The optical bridge sensor image data may therefore allow for real-time measurement of the image light provided to the eye box.

(48) Optical sensor **78** may be used to detect (measure) waveguide angle θ based on the image light **38B** emitted by projector **22B** (similar to optical sensor **40** of FIGS. 4 and 5). Optical sensor **78** may include one or more visible light optical sensors or cameras having one or more optical sensor pixels arranged in corresponding array(s) of optical sensor pixels. One or more lenses or microlenses may be disposed overlapping the optical sensor(s). Optical sensor **78** may include a quad cell sensor as one example. If desired, optical sensor **78** may also be used to detect (measure) optical misalignment between the left and right sides of the display (e.g., between waveguides **24A** and **24B**).

(49) Optical bridge sensor **74** may sometimes also be referred to as an optical misalignment detection sensor, an optical alignment sensor, or an optical misalignment detection module. If desired, optical bridge sensor **74** may be integrated within a sensor housing. The sensor housing may be formed from a part of main portion **18M** of housing **18** within nose bridge NB (FIG. 1), may be a separate housing enclosed within nose bridge NB of main portion **18M**, may be a frame or bracket that supports housing portion **18M**, or may be omitted. Optical bridge sensor **74** may have a first end mounted or coupled to waveguide **24B** and may have an opposing second end mounted or coupled to waveguide **24A** (e.g., using optically clear adhesive or other mounting structures).

(50) Output coupler **77B** may be formed from an output coupling prism or waveguide facets, mirror(s) (e.g., louvered mirrors), or diffractive grating structures such as a surface relief grating or volume holograms. In the example of FIG. 6, input coupler **28B**, output coupler **30B**, and output coupler **77B** are diffractive gratings in a layer of grating medium such as surface relief gratings (SRGs) in a layer of SRG medium. For example, output coupler **77B** may include a first SRG in a layer of SRG medium on waveguide **24B**, output coupler **30B** may include a second SRG in the layer of SRG medium on waveguide **24B**, and input coupler **28B** may include a third SRG in the layer of SRG medium on waveguide **24B**. This example is merely illustrative and, in general, input coupler **28B**, output coupler **30B**, and output coupler **77B** may be formed from any desired optical coupler structures integrated into or onto waveguide **24B** in any desired manner.

(51) Optical sensor **78** may generate optical sensor data in response to the image light **38B** coupled out of waveguide **24B** by output coupler **77B** in the direction of arrow **72**. However, this optical sensor data on its own may not be sufficient to detect waveguide angle θ . A reflective structure

such as mirror **82** may therefore be disposed overlapping optical sensor **78** at or on the side of waveguide **24B** opposite to optical bridge sensor **74** (e.g., waveguide **24B** may be interposed between mirror **82** and optical bridge sensor **74**). Mirror **82** may have a lateral (reflective) surface **86**. Output coupler **77B** may have a reflective diffractive mode or order that reflects (via diffraction) a first portion of the incident image light **38B** towards optical sensor **78** in the direction of arrow **72**. Output coupler **77B** may also have a transmissive diffractive mode or order that transmits (via diffraction) a second portion of the incident image light **38B** towards mirror **82** in the direction of arrow **74**. The reflective diffractive mode of output coupler **77B** may be configured to couple image light **38B** out of waveguide **24B** at the same angle with which output coupler **30B** couples image light **38B** out of waveguide (e.g., arrows **70** and **72** may be parallel).

(52) Mirror **82** reflects the second portion of image light **38B** incident in the direction of arrow **73** in the direction of arrow **76**. Arrow **76** may be oriented at a reflected angle with respect to arrow **73** (e.g., arrows **73** and **76** may be bisected by a reflective axis that is separated from arrow **73** and arrow **76** by equal angles). In situations where arrow **73** is oriented perpendicular to lateral surface **86** of mirror **82**, arrow **76** is also oriented perpendicular to lateral surface **86** of mirror **82**. Since output coupler **77B** is not Bragg-matched to the direction of arrow **76**, the second portion of the image light **38B** reflected by mirror **82** is transmitted through output coupler **77B** (without diffracting) and waveguide **24B** and is received at optical sensor **78**. Optical sensor **78** may generate optical sensor data based on the second portion of the image light **38B** reflected by mirror **82** and received in the direction of arrow **76**.

(53) Mirror **82** may be mounted to optical bridge sensor **74** using a mounting bracket or frame, may be mounted to waveguide **24B** using optically clear adhesive, and/or may be mounted to the housing for device **10**. In general, mirror **82** may be rigidly mounted such that lateral (reflective) surface **86** of mirror **82** extends parallel to a lateral (sensing) surface **84** of optical sensor **78** (e.g., parallel to the lateral surface **50** of projector **22B** and the X-axis of FIG. **6**) and such that any change in the position/orientation of optical sensor **78** over time will also produce the same change in the position/orientation of mirror **82**. In other words, mirror **82** and optical bridge sensor **74** may be mounted in device **10** in such a way as to form a corresponding fixed reference frame **80** for generating the optical sensor data.

(54) By mounting mirror **82** and optical bridge sensor **74** within device **10** in this way, the transmissive diffractive mode of output coupler **77B** may couple image light **38B** out of waveguide **24B** at an angle **20** (two times waveguide angle θ) with respect to the inverse of the angle with which output coupler **30B** couples image light **38B** out of waveguide **24B** (e.g., arrow **73** and arrow **70** may lie within respective lines that are oriented at angle **20** with respect to each other). The image light reflected by mirror **82** will therefore be oriented at angle **20** with respect to the image light reflected by the reflective diffractive mode of output coupler **77B** (e.g., arrow **76** may be oriented or tilted at angle **20** with respect to arrow **72**).

(55) The optical sensor data generated by optical sensor **78** will therefore include first optical sensor data generated in response to the image light **38B** reflected by output coupler **77B** and incident in the direction of arrow **72** and second optical sensor data generated in response to the image light **38B** reflected by mirror **82** and incident in the direction of arrow **76**. The control circuitry may process the optical sensor data to detect (e.g., identify, measure, compute, generate, etc.) waveguide angle θ . For example, the control circuitry may detect waveguide angle θ based on the lateral separation between the first optical sensor data generated from image light incident in the direction of arrow **72** and the second optical sensor data generated from image light incident in the direction of arrow **76** (e.g., the distance in pixel position between intensity peaks in a frame of optical sensor data generated by optical sensor **78**), since this lateral separation is expected to be zero when waveguide angle θ is equal to zero. If desired, the control circuitry may detect waveguide angle θ by comparing the optical sensor data generated by optical sensor **78** to calibration image data that indicates what the optical sensor data is expected to be under different

waveguide angles θ (e.g., as generated during manufacture or calibration of device **10**).

(56) If desired, mirror **82** may be replaced with an additional optical sensor such as optical sensor **90** in FIG. 7. As shown in FIG. 7, optical sensor **90** may be rigidly mounted within device **10** with respect to optical bridge sensor **74** (e.g., forming parting of reference frame **80**) at the location of mirror **82** of FIG. 6. Optical sensor **90** may have a lateral surface **91** (e.g., an image sensing surface) oriented parallel to lateral surface **84** of optical sensor **78**. Rather than reflecting the image light coupled out of waveguide **24B** by the transmissive diffractive mode of output coupler **77B**, optical sensor **90** may generate optical sensor data in response to the image light coupled out of waveguide **24B** by the transmissive diffractive mode of output coupler **77B** (e.g., as incident upon optical sensor **90** in the direction of arrow **73**). The control circuitry may then identify waveguide angle θ based on the optical sensor data gathered by optical sensor **90** (e.g., the position of the optical sensor data in the field of view of optical sensor **90**) and the optical sensor data gathered by optical sensor **78** (e.g., the position of the optical sensor data in the field of view of optical sensor **90**). The control circuitry may then mechanically adjust one or more components in device **10** and/or adjust the image data provided to projector **22B** to mitigate the optical effects at eye box **20B** of waveguide angle θ differing from a nominal or expected value (e.g., zero degrees).

(57) The examples of FIGS. 6 and 7 are merely illustrative. If desired, optical sensor **78** and/or optical sensor **90** may be infrared optical sensors and output coupler **77B** may couple infrared light emitted by projector **22B** (e.g., infrared light **64** of FIG. 5) out of waveguide **24B** and towards optical sensors **78** and/or **90**. The control circuitry may then use optical sensor data generated by optical sensors **78** and/or **90** to detect waveguide angle θ . Waveguide **24B** may also include a cross-coupler if desired. The optical sensors and components of FIGS. 4-7 may be combined if desired (e.g., device **10** may include two or more of optical sensor **40**, optical sensor **78**, mirror **82**, and optical sensor **90** for measuring waveguide angle and/or other alignments or distortions). Similar components and techniques may be used to detect the waveguide angle of waveguide **24A** (FIG. 2).

(58) FIG. 8 is a flow chart of illustrative operations involved in detecting and mitigating waveguide angle θ between waveguide **24B** and projector **22B**. Similar operations may also be used to detect changes in the waveguide angle between waveguide **24A** and projector **22A** (FIG. 2).

(59) At operation **100**, device **10** may project light into waveguide **24B** via input coupler **28B**. The light may include image light **38B** and/or infrared light **64** (FIG. 5). Waveguide **24B** may propagate the light via total internal reflection. Output coupler **30B** may couple a first portion of the light out of waveguide **24B** and towards eye box **20B**. If desired, a second portion of the light may be reflected off one or more lateral surfaces of waveguide **24B** and/or may be reflected by a zero-order mode of a diffractive grating in input coupler **28B** towards optical sensor **40** on projector **22B** (FIGS. 4 and 5). If desired, a third portion of the light may continue to propagate along waveguide **24B** without being coupled out of waveguide **24B** by output coupler **30B**. In these examples, output coupler **77B** (FIGS. 6 and 7) may couple a first portion of the third portion of the light out of the waveguide and towards optical sensor **78** in optical bridge sensor **74**. If desired, output coupler **77B** may also couple a second portion of the third portion of the light out of the waveguide and towards mirror **82** (FIG. 6) or optical sensor **90** (FIG. 7). In implementations where device **10** includes mirror **82**, mirror **82** may reflect the second portion of the third portion of the light towards optical sensor **78** in optical bridge sensor **74**.

(60) At operation **102**, device **10** may generate optical sensor data (e.g., image sensor data) based on the light projected by projector **22B**. For example, optical sensor **40** (FIGS. 4 and 5) may generate optical sensor data in response to the second portion of the light (e.g., portion **64'** of infrared light **64** of FIG. 5 or portion **38B'** of image light **38** of FIG. 4). Additionally or alternatively, optical sensor **78** may generate optical sensor data in response to the first portion of the third portion of the light (e.g., as shown by arrow **72** of FIGS. 6 and 7). Additionally or alternatively, optical sensor **78** or optical sensor **90** may generate optical sensor data in response to the second portion of the third portion of the light (e.g., as shown by arrows **73** and **76** of FIG. 6 or

arrow 73 of FIG. 7).

(61) At operation **104**, control circuitry in device **10** may detect (e.g., identify, measure, compute, calculate, generate, etc.) waveguide angle θ based on the gathered optical sensor data. The control circuitry may detect waveguide angle θ by comparing the gathered optical sensor data to calibrated or predetermined optical sensor data expected to be produced under one or more different (known) waveguide angles θ , as one example.

(62) At operation **106**, the control circuitry may take suitable action based on the detected waveguide angle. For example, the control circuitry may perform one or more operations to mitigate the detected waveguide angle. For example, the control circuitry may control one or more actuators or other mechanical adjustment structures to adjust the position of waveguide **24B**, the orientation of waveguide **24B**, the position of projector **22B**, or the strain applied to waveguide **24B** in a manner that reverses or mitigates the distortions produced on the image light by the detected amount of waveguide deformation. As another example, the control circuitry may predistort, warp, color-compensate, or otherwise adjust (e.g., digitally transform, translate, rotate, etc.) the image data provided to projector **22B** and used to produce image light **38B** in a manner that mitigates the effects of the detected waveguide angle deviating from a nominal or expected waveguide angle such as a waveguide angle of zero degrees. Additionally or alternatively, the control circuitry may issue an alert to the user or to a server identifying that waveguide angle exceeds a threshold value, may instruct the user to have some or all of device **10** repaired or replaced, and/or may instruct the user to mechanically adjust one or more components of device **10** or the position of device **10** on their head to mitigate the optical effects of changes in the waveguide angle. If desired, the optical sensor data may also be used to perform mechanical or digital adjustments to align the left projector, waveguide, and/or eye box with the right projector, waveguide, and/or eye box and/or to perform gaze tracking operations.

(63) The foregoing is merely illustrative and various modifications can be made to the described embodiments. The foregoing embodiments may be implemented individually or in any combination.

Claims

1. An electronic device comprising: a projector configured to emit image light; a waveguide; an input coupler configured to couple the image light into the waveguide; an output coupler configured to couple at least a portion of the image light out of the waveguide; an optical sensor configured to generate optical sensor data, wherein the optical sensor is mounted to the projector and faces the waveguide; and one or more processors configured to detect an angle between the projector and the waveguide based on the optical sensor data, wherein the waveguide has a first lateral surface and a second lateral surface opposite the first lateral surface, the waveguide is configured to propagate the image light via total internal reflection between the first and second lateral surfaces, the first lateral surface faces the projector, and the optical sensor is interposed between the first lateral surface and the projector.
2. The electronic device of claim 1, wherein the projector is configured to output infrared light that is different from the image light, the input coupler comprises a diffractive grating having a zero-order mode that configures the diffractive grating to reflect the infrared light from the projector back towards the optical sensor through the first lateral surface, and the optical sensor is configured to generate the optical sensor data based on the reflected infrared light.
3. The electronic device of claim 1, wherein the projector comprises an infrared emitter configured to emit the infrared light and wherein the infrared light and the image light are incident upon the input coupler through the first lateral surface of the waveguide.
4. The electronic device of claim 3, wherein the projector comprises: illumination optics configured to emit illumination light at visible wavelengths; and a reflective display panel configured generate

the image light by modulating the illumination light using image data, the reflective display panel being further configured to reflect the infrared light.

5. The electronic device of claim 4, wherein the projector comprises optics configured to combine the infrared light with the image light.

6. The electronic device of claim 5, wherein the infrared light emitted by the infrared emitter is incident upon a first side of the optics and wherein the illumination light emitted by the illumination optics is incident upon a second side of the optics.

7. The electronic device of claim 6, wherein the image light generated by the reflective display panel is incident upon a third side of the optics.

8. The electronic device of claim 7, wherein the optics are configured to output the image light and the infrared light through a fourth side of the optics.

9. The electronic device of claim 8, wherein the projector further comprises a lens that transmits the infrared light and the image light to the input coupler.

10. The electronic device of claim 1, wherein the optical sensor is mounted to a surface of the projector that faces the first lateral surface of the waveguide.

11. The electronic device of claim 1, wherein the input coupler comprises a first surface relief grating in a layer of surface relief grating medium on the waveguide, and the output coupler comprises a second surface relief grating in the layer of surface relief grating medium.

12. The electronic device of claim 1, further comprising: an optical bridge sensor that is different than the optical sensor and that is configured to measure the image light.

13. The electronic device of claim 1, wherein the one or more processors is configured to adjust image data provided to the projector based on the detected angle between the projector and the waveguide.

14. A method of operating a display system comprising: with a projector, emitting light into a waveguide through a first lateral surface of the waveguide; with an input coupler, coupling the light into the waveguide; with the waveguide, propagating the light via total internal reflection between the first lateral surface and a second lateral surface of the waveguide that is opposite the first lateral surface; with a first output coupler on the waveguide, coupling a first portion of the light out of the waveguide; with a second output coupler on the waveguide, coupling a second portion of the light out of the waveguide and towards an optical sensor through the first lateral surface; with the second output coupler, coupling a third portion of the light out of the waveguide and towards a mirror through the second lateral surface, wherein the waveguide is interposed between the optical sensor and the mirror; with the mirror, reflecting the third portion of the light towards the optical sensor through the second and first lateral surfaces of the waveguide; with the optical sensor, generating optical sensor data based on the second portion of the light and the reflected third portion of the light; and with one or more processors, detecting an angle between the projector and the waveguide based on the optical sensor data generated by the optical sensor.

15. The method of claim 14, wherein the first and second portions of the light comprise image light having visible wavelengths.

16. The method of claim 14, wherein the first portion of the light comprises image light having visible wavelengths and wherein the second portion of the light comprises infrared light.

17. The method of claim 16, further comprising: with the one or more processors, performing gaze tracking based on the infrared light.

18. The method of claim 14, wherein the second output coupler has a transmissive diffractive mode that is configured to diffract the third portion of the light and has a reflective diffractive mode that is configured to diffract the second portion of the light.

19. The method of claim 14, wherein a surface of the mirror extends parallel to a sensing surface of the optical sensor.

20. The method of claim 14, further comprising: with the one or more processors, adjusting image data provided to the projector based on the detected angle between the projector and the

waveguide.

21. A display comprising: a projector configured to generate light; a waveguide configured to propagate the light via total internal reflection between a first lateral surface and a second lateral surface opposite the first lateral surface; a first output coupler on the waveguide and configured to couple a first portion of the light out of the waveguide and towards an eye box; an optical sensor; a mirror, wherein the waveguide is interposed between the optical sensor and the mirror; and a second output coupler overlapping the optical sensor and the mirror, wherein the second output coupler is configured to couple a second portion of the light out of the waveguide and towards the mirror through the second lateral surface, and the second output coupler is configured to couple a third portion of the light out of the waveguide and towards the optical sensor through the first lateral surface.

22. The display of claim 21, wherein the mirror is configured to reflect the third portion of the light towards the optical sensor through the second lateral surface and the first lateral surface, the mirror extends parallel to a sensing surface of the optical sensor, the optical sensor is configured to generate sensor data based on the second and third portions of the light, and the display further comprises: processing circuitry configured to adjust the light based on the sensor data.
